1 The table shows some physical quantities.

Which row correctly identifies the quantities as scalars or vectors?

	acceleration	charge	kinetic energy	wavelength
A	scalar	vector	vector	scalar
B	vector	vector	scalar	scalar
C	scalar	scalar	scalar	vector
D	vector	scalar	scalar	scalar

2 Two quantities are measured.

$$L = 6.8 \pm 0.1\,\text{cm}$$

$$T = 2.42 \pm 0.08\,\text{s}$$

L and T are related to X by the equation shown.

$$X = \frac{4\pi^2 L}{T^2}$$

What is the calculated value and uncertainty of X?

A $45.8 \pm 0.2\,\text{cm}\,\text{s}^{-2}$

B $45.8 \pm 0.3\,\text{cm}\,\text{s}^{-2}$

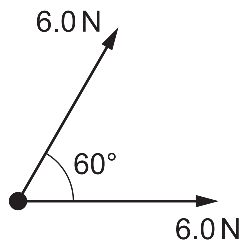
C $46 \pm 2\,\text{cm}\,\text{s}^{-2}$

D $46 \pm 4\,\text{cm}\,\text{s}^{-2}$

3 What are the SI base units of the watt?

A $\text{J}\,\text{s}^{-1}$ **B** $\text{kg}\,\text{m}^2\,\text{s}^{-3}$ **C** $\text{kg}\,\text{m}^2\,\text{s}^{-1}$ **D** $\text{N}\,\text{m}\,\text{s}^{-1}$

4 The diagram shows two forces of 6.0 N acting on an object. The angle between the lines of action of the two forces is 60°.



What is the magnitude of the resultant force?

A 6.0 N **B** 7.9 N **C** 10 N **D** 12 N